Pursuing the ecbench leads

373 fresh public single-target attempts; 336 verified scalar recoveries. Outcomes: {'error': 1, 'exhausted': 36, 'verified': 336}.

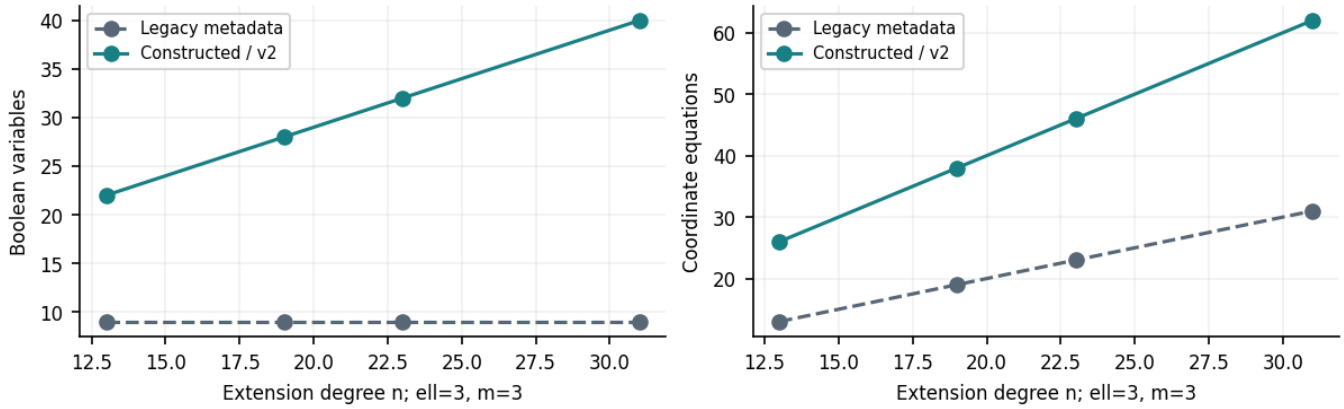
Both IC arms recovered all four held-out degree 37 public points, eight IC executions in total. Increasing the base from four to eight orbit columns changed mean counted S from 8.566326 to 2.618529; the four/eight count quotient is 3.271426. The eight-column/rho count quotient remains 9.012705. These are incomplete-cost diagnostics, not speedups.

The exact E0 degree 83 method now executes ordinary IC queries at both base sizes, but all four runs reached 256 queries without a point relation or rank gain. This discharges an implementation refusal; it does not pass the N83 target-recovery gate.

All session audits passed; unsuccessful deterministic outcomes replayed separately. The new method uses unknown factor-base logs and exact point witnesses at both integer widths. Versioned metadata includes all intermediate coordinates and equation blocks.

The exact E0 degree83 subgroup is retained. Bounded execution does not satisfy the N83 recovery gate. All ten mathematical questions remain partial. L0; no admitted wall time, fully priced speedup, asymptotic exponent or N131 transfer.

System shape: independently constructed inputs



ell=3, m=3. The plot uses native constructed counts and preserved legacy metadata; it is not a solving-degree or solver-cost claim.

Sources: PROTOCOL-v1.md; FINDINGS.md/findings.json; results.json; shape-controls.json; per-target.json; execution-manifest.json; parent literature/REPORT.md (29 references). The audit-reader amendment preserves the initial N83 reader failure and validates its correction against a pinned historical reader. See REPORT.md for source-linked run reports and exact remaining obligations.

Source-linked report and immutable evidence

Measured per-curve diagnostics

Verified-only means; ratios are missing on censored cells. Native work unpriced: these are count quotients, not full-cost ratios. Every target and failure remains in per-target.json.

Curve	Arm	Verified	Mean S	Count / rho
icv1-f2m13-t181-515ee569	f6-ic-v1	0/2	—	—
icv1-f2m13-t181-515ee569	f6-ic-v2	0/2	—	—
icv1-f2m13-t181-515ee569	inherited-f4-v1	0/2	—	—
icv1-f2m13-t181-515ee569	inherited-f4-v2	0/2	—	—
icv1-f2m13-t181-515ee569	rho	2/2	14.70939	1.00000
icv1-f2m13-t181-515ee569	rho-aa	2/2	14.70939	1.00000
icv1-f2m19-t797-b6cf2467	f6-ic-v1	0/2	—	—
icv1-f2m19-t797-b6cf2467	f6-ic-v2	0/2	—	—
icv1-f2m19-t797-b6cf2467	inherited-f4-v1	0/2	—	—
icv1-f2m19-t797-b6cf2467	inherited-f4-v2	0/2	—	—
icv1-f2m19-t797-b6cf2467	rho	2/2	3.04449	1.00000
icv1-f2m19-t797-b6cf2467	rho-aa	2/2	3.04449	1.00000
icv1-f2m23-t5197-69e76b73	f6-ic-v1	0/2	—	—
icv1-f2m23-t5197-69e76b73	f6-ic-v2	0/2	—	—
icv1-f2m23-t5197-69e76b73	inherited-f4-v1	0/2	—	—
icv1-f2m23-t5197-69e76b73	inherited-f4-v2	0/2	—	—
icv1-f2m23-t5197-69e76b73	rho	2/2	1.02746	1.00000
icv1-f2m23-t5197-69e76b73	rho-aa	2/2	1.02746	1.00000
icv1-f2m31-tm90707-c95f16f5	f6-ic-v1	0/2	—	—
icv1-f2m31-tm90707-c95f16f5	f6-ic-v2	0/2	—	—
icv1-f2m31-tm90707-c95f16f5	inherited-f4-v1	0/2	—	—
icv1-f2m31-tm90707-c95f16f5	inherited-f4-v2	0/2	—	—
icv1-f2m31-tm90707-c95f16f5	rho	2/2	1.14982	1.00000
icv1-f2m31-tm90707-c95f16f5	rho-aa	2/2	1.14982	1.00000
icv1-f2m37-tm534059-32aad96b	orbit-c4	4/4	8.56633	29.48440
icv1-f2m37-tm534059-32aad96b	orbit-c8	4/4	2.61853	9.01271
icv1-f2m37-tm534059-32aad96b	rho	4/4	0.29054	1.00000
icv1-f2m37-tm534059-32aad96b	rho-aa	4/4	0.29054	1.00000
icv1-f2m13-t181-515ee569	exact-c4	4/4	127.71789	8.90943
icv1-f2m13-t181-515ee569	existing-orbit-c4	4/4	19.78156	1.37994
icv1-f2m13-t181-515ee569	orbit-c4	4/4	14.78609	1.03146
icv1-f2m13-t181-515ee569	rho	4/4	14.33513	1.00000
icv1-f2m13-t181-515ee569	rho-aa	4/4	14.33513	1.00000
icv1-f2m19-t797-b6cf2467	exact-c4	4/4	33.99111	10.55166
icv1-f2m19-t797-b6cf2467	existing-orbit-c4	4/4	4.07241	1.26417
icv1-f2m19-t797-b6cf2467	orbit-c4	4/4	3.54168	1.09942
icv1-f2m19-t797-b6cf2467	rho	4/4	3.22140	1.00000
icv1-f2m19-t797-b6cf2467	rho-aa	4/4	3.22140	1.00000
icv1-f2m23-t5197-69e76b73	exact-c4	4/4	13.71135	12.04707
icv1-f2m23-t5197-69e76b73	existing-orbit-c4	4/4	1.89964	1.66906
icv1-f2m23-t5197-69e76b73	orbit-c4	4/4	2.46321	2.16422
icv1-f2m23-t5197-69e76b73	rho	4/4	1.13815	1.00000
icv1-f2m23-t5197-69e76b73	rho-aa	4/4	1.13815	1.00000
icv1-f2m31-tm90707-c95f16f5	exact-c4	4/4	26.77523	22.38191
icv1-f2m31-tm90707-c95f16f5	existing-orbit-c4	4/4	2.12457	1.77597
icv1-f2m31-tm90707-c95f16f5	orbit-c4	4/4	1.86664	1.56036
icv1-f2m31-tm90707-c95f16f5	rho	4/4	1.19629	1.00000
icv1-f2m31-tm90707-c95f16f5	rho-aa	4/4	1.19629	1.00000
icv1-fp16-t295-8d3c3165	abs128-mitm2	8/8	75.95687	32.82865
icv1-fp16-t295-8d3c3165	abs16-mitm2	8/8	11.02524	4.76512
icv1-fp16-t295-8d3c3165	abs32-mitm2	8/8	10.34294	4.47023
icv1-fp16-t295-8d3c3165	abs64-mitm2	8/8	22.92477	9.90811
icv1-fp16-t295-8d3c3165	abs64-subtract3	8/8	23.72042	10.25199

Curve	Arm	Verified	Mean S	Count / rho
icv1-fp16-t295-8d3c3165	rho	8/8	2.31374	1.00000
icv1-fp16-t295-8d3c3165	rho-aa	8/8	2.31374	1.00000
icv1-fp18-tm337-d28d5e09	abs128-mitm2	8/8	38.88323	25.48853
icv1-fp18-tm337-d28d5e09	abs16-mitm2	8/8	14.32550	9.39058
icv1-fp18-tm337-d28d5e09	abs32-mitm2	8/8	10.50614	6.88693
icv1-fp18-tm337-d28d5e09	abs64-mitm2	8/8	13.73726	9.00498
icv1-fp18-tm337-d28d5e09	abs64-subtract3	8/8	18.99209	12.44960
icv1-fp18-tm337-d28d5e09	rho	8/8	1.52552	1.00000
icv1-fp18-tm337-d28d5e09	rho-aa	8/8	1.52552	1.00000
icv1-fp20-t727-cd198a38	abs128-mitm2	8/8	25.29264	17.22429
icv1-fp20-t727-cd198a38	abs16-mitm2	8/8	22.58380	15.37957
icv1-fp20-t727-cd198a38	abs32-mitm2	8/8	11.01955	7.50431
icv1-fp20-t727-cd198a38	abs64-mitm2	8/8	9.74397	6.63564
icv1-fp20-t727-cd198a38	abs64-subtract3	8/8	21.59557	14.70658
icv1-fp20-t727-cd198a38	rho	8/8	1.46843	1.00000
icv1-fp20-t727-cd198a38	rho-aa	8/8	1.46843	1.00000
icv1-fp22-tm1385-475dcb5f	abs128-mitm2	8/8	15.13896	16.14176
icv1-fp22-tm1385-475dcb5f	abs16-mitm2	8/8	37.40202	39.87950
icv1-fp22-tm1385-475dcb5f	abs32-mitm2	8/8	18.49377	19.71879
icv1-fp22-tm1385-475dcb5f	abs64-mitm2	8/8	10.81144	11.52758
icv1-fp22-tm1385-475dcb5f	abs64-subtract3	8/8	38.58922	41.14535
icv1-fp22-tm1385-475dcb5f	rho	8/8	0.93788	1.00000
icv1-fp22-tm1385-475dcb5f	rho-aa	8/8	0.93788	1.00000
icv1-f2m83-tm6151469093347-debefd74	legacy-ic	0/1	—	—
icv1-f2m83-tm6151469093347-debefd74	orbit-c16	0/2	—	—
icv1-f2m83-tm6151469093347-debefd74	orbit-c4	0/2	—	—

Original questions and remaining gates

Q1. Prime-field base-size balance tested on fresh targets; scalable cheap decomposition/full cost open.

Q2. Actual chained metadata corrected and checked; solving-degree growth open.

Q3. Enumerated quotient witnesses lift exactly; compact algebraic correspondence open.

Q4. Parent nonlinear-base evidence retained; scalable membership/full cost open.

Q5. Parent quasi-subfield census retained; useful general family/full IC open.

Q6. Intermediate-variable accounting corrected; higher-arity solver/lifting advantage open.

Q7. Width-generic modular rank implemented; general useful-rank and descent law open.

Q8. Paid certified isogeny/Hecke transport open.

Q9. General norm and cheaper elliptic-component solver open.

Q10. Successful N83 recovery, full costs, isolated independent hardware and growing-size crossover open.

No new mathematical theorem. Native Rust sources and sealed records derive all values; this Python file only draws and lays out them. PR1853 remains subject to review and applicable CI. No research jobs continue after this completed local campaign.